

Plan Model — Complete Equation

Pure Agronomic Decision Engine — Sentinel-Grade v3 + Land Activity Classifier

AgroCloud · Satellite Intelligence · Crop Alert Pipeline · 2026

Core Principle

Agronomic Truth = Spectral Indices (NDVI, NDMI, NDWI)
+ Temporal Deltas (Δ)
+ Spatial Distribution (Histogram)
+ Agricultural Context (Land Activity)

Governing constraint: One decision = **one dominant signal** (no averaging, no multi-label output).

1) INPUT MODEL

```
{
  "current": { "NDVI": [-1,1], "NDMI": [-1,1], "NDWI": [-1,1] },
  "history": { "prev1": {...}, "prev2": {...} },
  "histogram": {
    "bins": [{ "class": "healthy|stressed|bare", "ratio": float }]
  }
}
```

Data pipeline:

Sentinel-2 L2A → Statistical API → zonal mean (current, prev1, prev2)
→ NDVI histogram bins
→ Land Activity history (3 scenes NDVI)

2) PRE-COMPUTATION LAYERS

Layer 0 — Land Activity Classifier (mandatory before any alert)

Priority	Condition	land_state
1	NDVI < 0.10	BARE_SOIL_UNPLANTED
2	NDVI < 0.12 AND NDMI ≤ 0.08 AND NDWI ≤ 0.08	BARE_SOIL_UNPLANTED
3	NDVI < 0.15 AND ΔNDVI ≤ 0.02 (×2 scenes)	NON_CULTIVATED_STABLE
4	NDVI > 0.20 AND previous baseline < 0.15	ACTIVE_CROP_ESTABLISHED
5	NDVI ≥ 0.20	ACTIVE_CROP_ONGOING
6	previous canopy ≥ 0.20	ACTIVE_CROP_ONGOING
else	—	NON_CULTIVATED_STABLE

```
NO_CROP = { BARE_SOIL_UNPLANTED, NON_CULTIVATED_STABLE }

HEALTHY_allowed ⇔ NDVI > 0.40 AND (ΔNDVI1 > 0 OR ΔNDVI2 > 0)
                    AND land_state ∉ NO_CROP
```

Layer 1 — Delta & Primary Signal Lock

```
ΔNDVI = current.NDVI - prev1.NDVI
ΔNDMI = current.NDMI - prev1.NDMI
ΔNDWI = current.NDWI - prev1.NDWI

IF |ΔNDVI| > |ΔNDMI| AND |ΔNDVI| > |ΔNDWI| → PRIMARY = VEGETATION
IF |ΔNDMI| > |ΔNDVI| AND |ΔNDMI| > |ΔNDWI| → PRIMARY = MOISTURE
IF |ΔNDWI| > |ΔNDVI| AND |ΔNDWI| > |ΔNDMI| → PRIMARY = SURFACE_WATER
(tie-break: VEGETATION > MOISTURE > SURFACE_WATER)
```

Layer 2 — Trend Engine

```
TrendScore = (ΔNDVI × 0.6) + (ΔNDMI × 0.3) + (ΔNDWI × 0.1)

IF TrendScore < -0.025 → DECLINING
IF TrendScore > +0.025 → IMPROVING
ELSE → STABLE
```

Layer 3 — Water Engine (tight thresholds)

water_status	Condition
SEVERE_DROUGHT	$NDMI \leq -0.12$ AND $NDWI \leq 0.03$
MODERATE_DROUGHT	$-0.12 < NDMI \leq -0.06$
WATERLOGGING	$NDMI \geq 0.25$ AND $NDWI \geq 0.30$
NORMAL	otherwise

Layer 4 — Vegetation Engine (non-overlapping bins)

vegetation_status	NDVI
DEGRADED	≤ 0.18
STRESSED	$0.18 < NDVI \leq 0.35$
HEALTHY	$0.35 < NDVI \leq 0.65$
VERY_HEALTHY	> 0.65

IF land_state ∈ NO_CROP → vegetation = DEGRADED
IF (HEALTHY|VERY_HEALTHY) AND NOT HEALTHY_allowed → downgrade to STRESSED/DEGRADED

Layer 5 — Biotic vs Environmental

IF $\Delta NDVI < 0$ AND $|\Delta NDMI| < 0.015$ → BIOTIC_STRESS
IF $\Delta NDVI < 0$ AND $\Delta NDMI < 0$ AND $\Delta NDWI < 0$ → ENVIRONMENTAL_STRESS
ELSE → NONE

Layer 6 — Spatial Histogram Engine

healthy_ratio = Σ (bins where class = healthy)
stress_ratio = Σ (bins where class = stressed OR bare)
entropy_proxy = $1 - \max(\text{all bin ratios})$

HIGH_ALERT ⇔ stress_ratio ≥ 0.45
ALERT_RISK ⇔ healthy_ratio ≤ 0.45
STABLE_SPATIAL ⇔ NOT HIGH_ALERT AND NOT ALERT_RISK

3) DECISION ENGINE (Strict Priority — NO TIES)

```
Input → Land Activity Classifier
      ↓ land_state ∈ NO_CROP?
YES → NO_CROP_ACTIVITY_DETECTED | risk=LOW | show_crop_health_alert=false
NO → Compute Δ, Primary Signal, Trend, Water, Vegetation, Biotic, Spatial
      ↓ SEVERE_DROUGHT? → IRRIGATION_REQUIRED_IMMEDIATELY
      ↓ WATERLOGGING? → DRAINAGE_REQUIRED
      ↓ DEGRADED + BIOTIC_STRESS? → DISEASE_INSPECTION_REQUIRED
      ↓ trend = DECLINING? → EARLY_WARNING_MONITORING
      ↓ else → FIELD_STABLE
```

4) RISK ENGINE (Forced Separation)

```
CRITICAL ⇔ SEVERE_DROUGHT
          OR WATERLOGGING
          OR (DEGRADED + BIOTIC_STRESS)

HIGH      ⇔ STRESSED + DECLINING

MEDIUM    ⇔ MODERATE_DROUGHT
          OR ALERT_RISK
          OR (default fallback)

LOW        ⇔ land_state ∈ NO_CROP
          OR (HEALTHY|VERY_HEALTHY) + (STABLE|IMPROVING)
```

5) CONFIDENCE ENGINE (Anti-Uniform)

```
signal_strength      = clamp( max(|ΔNDVI|, |ΔNDMI|, |ΔNDWI|) × 3 , 0, 1 )
spatial_separation    = clamp( 1 - entropy_proxy , 0, 1 )
temporal_consistency = 1   if sign(ΔNDVI)=sign(ΔNDMI)=sign(ΔNDWI)
                      = 0.5 otherwise

confidence = clamp(
  0.5 × signal_strength
+ 0.3 × spatial_separation
+ 0.2 × temporal_consistency
, 0, 1)
```

6) MAP PRESENTATION MODEL (Crop Alert Beacons)

State	Color	Pulse
Bare / unplanted (NO_CROP)	#90a4ae / #78909c gray	No
NDVI < 0.05	Red	Based on ΔNDVI
0.05–0.25	Orange	Based on ΔNDVI
0.25–0.40	Yellow	Based on ΔNDVI
0.40–0.60	Light green	Based on ΔNDVI
0.60–0.75	Green	Based on ΔNDVI
≥ 0.75	Dark green	Based on ΔNDVI

```
ΔNDVI_scene = scene[0] - scene[1]
ALERT_RED   ⇔ ΔNDVI ≤ -0.15
ALERT_GREEN ⇔ ΔNDVI ≥ +0.15
```

7) OUTPUT MODEL (Strict JSON)

```
{
  "land_state": "BARE_SOIL_UNPLANTED | NON_CULTIVATED_STABLE | ACTIVE_CROP_ESTABLISHED |
ACTIVE_CROP_ONGOING",
  "show_crop_health_alert": true,
  "vegetation_status": "VERY_HEALTHY | HEALTHY | STRESSED | DEGRADED",
  "water_status": "SEVERE_DROUGHT | MODERATE_DROUGHT | WATERLOGGING | NORMAL",
  "biotic_status": "BIOTIC_STRESS | ENVIRONMENTAL_STRESS | NONE",
  "trend": "IMPROVING | DECLINING | STABLE",
  "primary_signal": "VEGETATION | MOISTURE | SURFACE_WATER",
  "agronomic_decision": "NO_CROP_ACTIVITY_DETECTED | IRRIGATION_REQUIRED_IMMEDIATELY | ...",
  "risk_level": "LOW | MEDIUM | HIGH | CRITICAL",
  "confidence": 0.00,
  "explanation": { "primary_driver", "secondary_driver", "spatial_signal" }
}
```

8) Master Equation

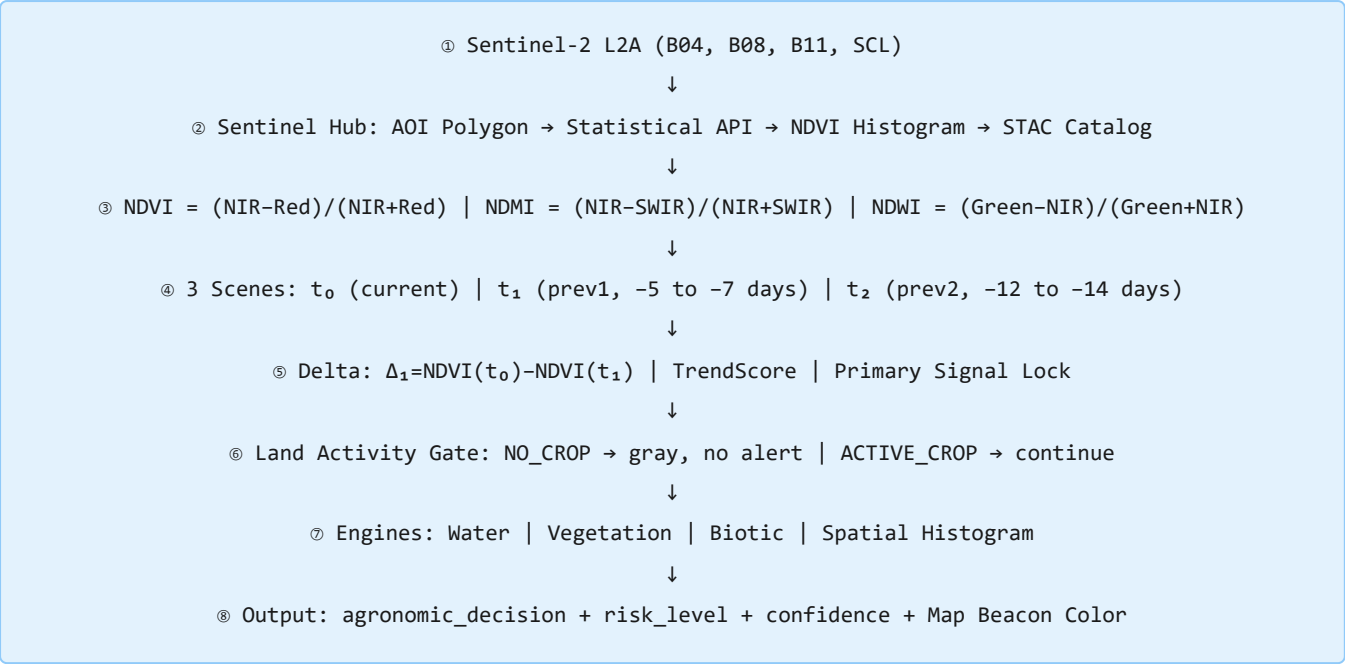
```
Decision = f_land(land_state)
  ◦ f_priority(water, vegetation, biotic, trend)
  ◦ f_risk(water, vegetation, biotic, trend, spatial)
  ◦ f_confidence( $\Delta$ , spatial)
  ◦ f_map(NDVI, land_state,  $\Delta$ NDVI_scene)
```

WHERE:

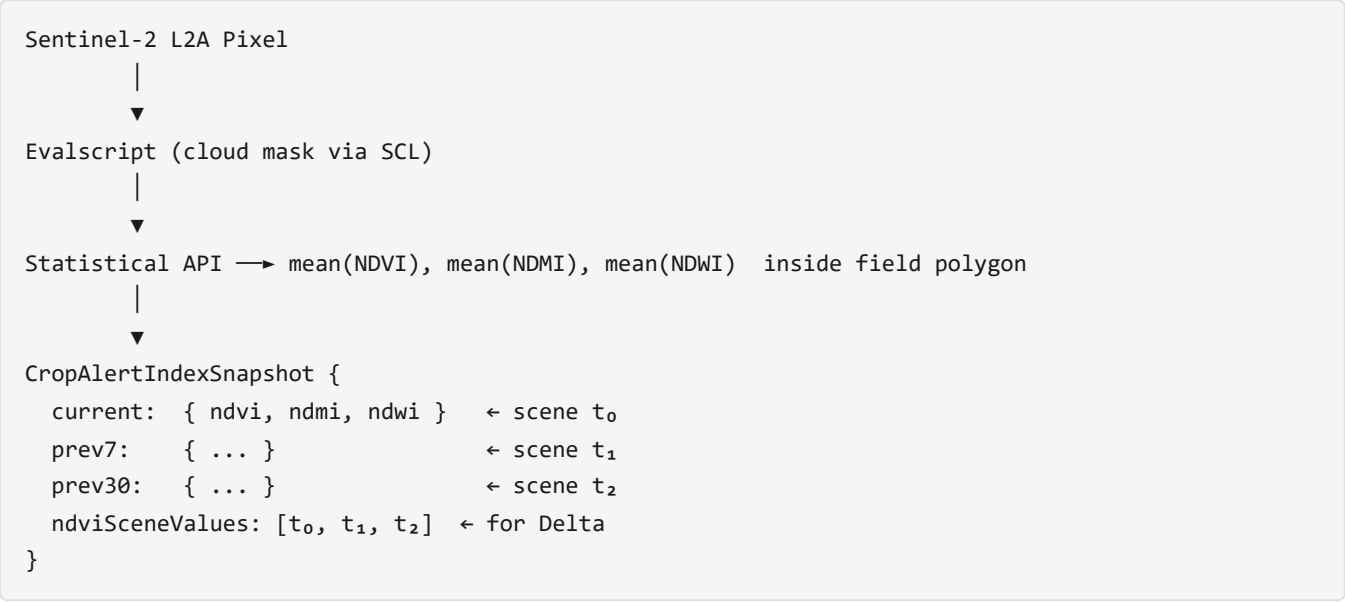
```
f_land      = NO_CROP → suppress health alerts, gray map, LOW risk
f_priority  = strict ordered rules (no averaging)
f_risk      = forced tier separation
f_confidence= weighted non-uniform score
f_map       = NDVI ramp IF active crop ELSE neutral gray
```

Equation Flow: Satellite → Delta → Alert

End-to-End Pipeline



① From Satellite to Numbers



Files: sentinelHubStatisticsApi.ts → siCropAlertSentinelLive.ts

② Delta Processing (Core Equation)

A) Spectral delta (current – prev1)

$$\Delta\text{NDVI} = \text{NDVI}(t_0) - \text{NDVI}(t_1)$$

$$\Delta\text{NDMI} = \text{NDMI}(t_0) - \text{NDMI}(t_1)$$

$$\Delta\text{NDWI} = \text{NDWI}(t_0) - \text{NDWI}(t_1)$$

B) Temporal delta across 3 scenes

$$\Delta_1 = \text{NDVI}(t_0) - \text{NDVI}(t_1)$$

$$\Delta_2 = \text{NDVI}(t_1) - \text{NDVI}(t_2)$$

$$\text{Growth Pattern} = (\Delta_1 > 0) \text{ OR } (\Delta_2 > 0)$$

$$\text{Flat Pattern} = |\Delta_1| \leq 0.02 \text{ AND } |\Delta_2| \leq 0.02$$

C) Trend Score

$$\text{TrendScore} = \Delta\text{NDVI} \times 0.6 + \Delta\text{NDMI} \times 0.3 + \Delta\text{NDWI} \times 0.1$$

$$< -0.025 \rightarrow \text{DECLINING} \mid > +0.025 \rightarrow \text{IMPROVING} \mid \text{else} \rightarrow \text{STABLE}$$

D) Primary Signal Lock

$$\max(|\Delta\text{NDVI}|, |\Delta\text{NDMI}|, |\Delta\text{NDWI}|) \rightarrow \text{VEGETATION} \mid \text{MOISTURE} \mid \text{SURFACE_WATER}$$

E) Map temporal alert badge

$$\Delta\text{NDVI_scene} = \text{ndviSceneValues}[0] - \text{ndviSceneValues}[1]$$

$$\Delta\text{NDVI_scene} \leq -0.15 \rightarrow \text{ALERT_RED} \mid \geq +0.15 \rightarrow \text{ALERT_GREEN}$$

③ Land Activity Gate

$$\text{NDVI}(t_0, t_1, t_2) + \text{NDMI} + \text{NDWI}$$

$$\rightarrow \text{NDVI} < 0.10? \rightarrow \text{BARE_SOIL_UNPLANTED} \rightarrow \text{STOP (NO_CROP, gray)}$$

$$\rightarrow \text{NDVI} < 0.15 \text{ AND flat } \Delta? \rightarrow \text{NON_CULTIVATED_STABLE} \rightarrow \text{STOP}$$

$$\rightarrow \text{NDVI} > 0.20 \text{ after low baseline?} \rightarrow \text{ACTIVE_CROP_ESTABLISHED} \rightarrow \text{GO}$$

$$\rightarrow \text{NDVI} \geq 0.20 \text{ OR had canopy?} \rightarrow \text{ACTIVE_CROP_ONGOING} \rightarrow \text{GO}$$

④ Engines → Decision (Strict Priority)

ORDER (first match wins):

- | | |
|-----------------------------|-----------------------------------|
| 1. land_state ∈ NO_CROP | → NO_CROP_ACTIVITY_DETECTED |
| 2. SEVERE_DROUGHT | → IRRIGATION_REQUIRED_IMMEDIATELY |
| 3. WATERLOGGING | → DRAINAGE_REQUIRED |
| 4. DEGRADED + BIOTIC_STRESS | → DISEASE_INSPECTION_REQUIRED |
| 5. trend = DECLINING | → EARLY_WARNING_MONITORING |
| 6. else | → FIELD_STABLE |

⑤ Risk + Confidence

```
CRITICAL ← SEVERE_DROUGHT | WATERLOGGING | (DEGRADED+BIOTIC)
HIGH      ← STRESSED + DECLINING
MEDIUM   ← MODERATE_DROUGHT | ALERT_RISK
LOW       ← NO_CROP | (HEALTHY+STABLE/IMPROVING)
```

```
confidence = 0.5×signal + 0.3×spatial + 0.2×temporal
```

⑥ Code Pipeline (SatelliteIntelligence)

```
Agro Structures Fields (geometry)
  |
  ▼
fetchCropAlertSentinelLiveBatch() ← Statistical API per field
  |
  ▼
buildSnapshotsFromSentinelSeries() ← current + prev + ndviSceneValues[3]
  |
  ▼
runCropAlertEngine()               ← NDVI decision + status
  |
  ▼
resolveUnifiedFieldPresentation() ← Land Activity + Delta + Color
  |
  ▼
SiCropAlertMapMarkersLayer        ← beacons on map
```

One-Line Equation

```
Alert = MapColor( NDVI(t0) )
  ◦ Decision( Δ(t0,t1), TrendScore, Water, Vegetation, Biotic, Spatial )
  ◦ LandGate( NDVI(t0,t1,t2), NDMI, NDWI )
  ◦ SentinelMean( Field_AOI, Scene_Dates )
  ◦ Sentinel2_L2A
```

9) Implementation Files

Layer	File
Land Activity	frontend/src/lib/landActivityClassifier.ts
Decision Engine v3	frontend/src/lib/pureAgronomicDecisionEngine.ts
Map Color + Zones	frontend/src/lib/siCropAlertNdviZones.ts
Sentinel Data	sentinelHubStatisticsApi.ts + siCropAlertSentinelLive.ts
Map Beacons	SiCropAlertMapMarker.tsx + SiCropAlertMapMarkersLayer.tsx